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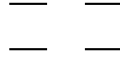
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1 Orbitals And Their Symmetry

We use the following ordering of orbitals here (assuming degeneracy of LUMOs and HOMOs):

$$\begin{array}{cc} b_2^* & \text{---} & \text{---} & a_2^* \\ a_2 & \text{---} & \text{---} & b_2 \end{array}$$

In the following, we will leave out the explicit labeling by orbital symmetry. Each monomer will simply be written as:



where the character of the orbital is defined by its position.

Dimer configurations will be written as their monomer occupations, by writing one monomer to the left and one to the right. The orbital ordering within the group of monomer orbitals follows the above mentioned definition.

The transformation of monomer orbitals into dimer orbitals is given by knowing the negative and positive linear combinations of monomer orbitals into dimer orbitals. Transforming this known equation system leads to:

$$\begin{aligned} a_2^{*l} &= +b_1^* + a_2^* \\ b_2^{*l} &= +a_1^* + b_2^* \\ a_2^l &= +b_1 + a_2 \\ b_2^l &= +a_1 + b_2 \\ a_2^{*r} &= -b_1^* + a_2^* \\ b_2^{*r} &= -a_1^* + b_2^* \\ a_2^r &= -b_1 + a_2 \\ b_2^r &= -a_1 + b_2 \end{aligned}$$

2 Monomer States and Configurations

$$\begin{aligned}
S &= + \begin{pmatrix} \overline{} & \overline{} \\ \downarrow\uparrow & \downarrow\uparrow \end{pmatrix} = \left| \underbrace{(b_2^l)^2 (b_2^{*l})^0 (a_2^l)^2 (a_2^{*l})^0}_{a_1} \right| \\
Q &= + \begin{pmatrix} \uparrow & \uparrow \\ \uparrow & \uparrow \end{pmatrix} = \left| \underbrace{(b_2^l)^1 (b_2^{*l})^1 (a_2^l)^1 (a_2^{*l})^1}_{a_1} \right| \\
i^3 a_1 &= + \begin{pmatrix} \uparrow & \overline{} \\ \downarrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} \overline{} & \uparrow \\ \uparrow & \downarrow\uparrow \end{pmatrix} = \left| \underbrace{(b_2^l)^1 (b_2^{*l})^1 (a_2^l)^2 (a_2^{*l})^0}_{a_1} \right| - \left| \underbrace{(b_2^l)^2 (b_2^{*l})^0 (a_2^l)^1 (a_2^{*l})^1}_{a_1} \right| \\
e^3 a_1 &= + \begin{pmatrix} \uparrow & \overline{} \\ \downarrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} \overline{} & \uparrow \\ \uparrow & \downarrow\uparrow \end{pmatrix} = \left| \underbrace{(b_2^l)^1 (b_2^{*l})^1 (a_2^l)^2 (a_2^{*l})^0}_{a_1} \right| + \left| \underbrace{(b_2^l)^2 (b_2^{*l})^0 (a_2^l)^1 (a_2^{*l})^1}_{a_1} \right| \\
i^3 b_1 &= + \begin{pmatrix} \uparrow & \overline{} \\ \uparrow & \downarrow\uparrow \end{pmatrix} - \begin{pmatrix} \overline{} & \uparrow \\ \downarrow\uparrow & \uparrow \end{pmatrix} = \left| \underbrace{(b_2^l)^2 (b_2^{*l})^1 (a_2^l)^1 (a_2^{*l})^0}_{b_1} \right| - \left| \underbrace{(b_2^l)^1 (b_2^{*l})^0 (a_2^l)^2 (a_2^{*l})^1}_{b_1} \right| \\
e^3 b_1 &= + \begin{pmatrix} \uparrow & \overline{} \\ \uparrow & \downarrow\uparrow \end{pmatrix} + \begin{pmatrix} \overline{} & \uparrow \\ \downarrow\uparrow & \uparrow \end{pmatrix} = \left| \underbrace{(b_2^l)^2 (b_2^{*l})^1 (a_2^l)^1 (a_2^{*l})^0}_{b_1} \right| + \left| \underbrace{(b_2^l)^1 (b_2^{*l})^0 (a_2^l)^2 (a_2^{*l})^1}_{b_1} \right|
\end{aligned}$$

3 Linear Combinations

3.1 $S * Q + Q * S$

$$S * Q + Q * S$$

$$+ \begin{pmatrix} \overline{\quad} & \overline{\quad} & \uparrow & \uparrow \\ \uparrow\uparrow & \uparrow\uparrow & \uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & \uparrow & \overline{\quad} & \overline{\quad} \\ \uparrow & \uparrow & \uparrow\uparrow & \uparrow\uparrow \end{pmatrix}$$

monomer ci vectors:

$$+ \left| \underbrace{(b_2^l)^2 (b_2^{*l})^0 (a_2^l)^2 (a_2^{*l})^0}_{a_1^l} \right| \cdot \left| \underbrace{(b_2^r)^1 (b_2^{*r})^1 (a_2^r)^1 (a_2^{*r})^1}_{a_1^r} \right| + \left| \underbrace{(b_2^l)^1 (b_2^{*l})^1 (a_2^l)^1 (a_2^{*l})^1}_{a_1^l} \right| \cdot \left| \underbrace{(b_2^r)^2 (b_2^{*r})^0 (a_2^r)^2 (a_2^{*r})^0}_{a_1^r} \right|$$

substitution by dimer orbitals:

$$+ \left| \underbrace{(+a_1 + b_2)^2 (+a_1^* + b_2^*)^0 (+b_1 + a_2)^2 (+b_1^* + a_2^*)^0}_{a_1^l} \right| \cdot \left| \underbrace{(-a_1 + b_2)^1 (-a_1^* + b_2^*)^1 (-b_1 + a_2)^1 (-b_1^* + a_2^*)^1}_{a_1^r} \right| + \dots$$

(multiplied out and summed up) dimer ci vectors:

$$\begin{aligned} & -2 \cdot \left| \underbrace{(a_1)^2 (b_2)^1 (b_1)^2 (a_2)^1 (a_1^*)^0 (b_2^*)^1 (b_1^*)^0 (a_2^*)^1}_{a_1} \right| - 2 \cdot \left| \underbrace{(a_1)^2 (b_2)^1 (b_1)^1 (a_2)^2 (a_1^*)^0 (b_2^*)^1 (b_1^*)^1 (a_2^*)^0}_{a_1} \right| \\ & -2 \cdot \left| \underbrace{(a_1)^2 (b_2)^1 (b_1)^2 (a_2)^1 (a_1^*)^1 (b_2^*)^0 (b_1^*)^1 (a_2^*)^0}_{a_1} \right| - 2 \cdot \left| \underbrace{(a_1)^2 (b_2)^1 (b_1)^1 (a_2)^2 (a_1^*)^1 (b_2^*)^0 (b_1^*)^0 (a_2^*)^1}_{a_1} \right| \\ & -2 \cdot \left| \underbrace{(a_1)^1 (b_2)^2 (b_1)^2 (a_2)^1 (a_1^*)^0 (b_2^*)^1 (b_1^*)^1 (a_2^*)^0}_{a_1} \right| - 2 \cdot \left| \underbrace{(a_1)^1 (b_2)^2 (b_1)^1 (a_2)^2 (a_1^*)^0 (b_2^*)^1 (b_1^*)^0 (a_2^*)^1}_{a_1} \right| \\ & -2 \cdot \left| \underbrace{(a_1)^1 (b_2)^2 (b_1)^2 (a_2)^1 (a_1^*)^1 (b_2^*)^0 (b_1^*)^0 (a_2^*)^1}_{a_1} \right| - 2 \cdot \left| \underbrace{(a_1)^1 (b_2)^2 (b_1)^1 (a_2)^2 (a_1^*)^1 (b_2^*)^0 (b_1^*)^1 (a_2^*)^0}_{a_1} \right| \end{aligned}$$

3.2 $S * Q - Q * S$

$$S * Q - Q * S$$

$$+ \begin{pmatrix} \overline{\quad} & \overline{\quad} & \uparrow\uparrow & \uparrow\uparrow \\ \uparrow\uparrow & \uparrow\uparrow & \uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} \uparrow\uparrow & \uparrow\uparrow & \overline{\quad} & \overline{\quad} \\ \uparrow & \uparrow & \uparrow\uparrow & \uparrow\uparrow \end{pmatrix}$$

monomer ci vectors:

$$+ \left| \underbrace{(b_2^l)^2 (b_2^{*l})^0 (a_2^l)^2 (a_2^{*l})^0}_{a_1^l} \right| \cdot \left| \underbrace{(b_2^r)^1 (b_2^{*r})^1 (a_2^r)^1 (a_2^{*r})^1}_{a_1^r} \right| - \left| \underbrace{(b_2^l)^1 (b_2^{*l})^1 (a_2^l)^1 (a_2^{*l})^1}_{a_1^l} \right| \cdot \left| \underbrace{(b_2^r)^2 (b_2^{*r})^0 (a_2^r)^2 (a_2^{*r})^0}_{a_1^r} \right|$$

substitution by dimer orbitals:

$$+ \left| \underbrace{(+a_1 + b_2)^2 (+a_1^* + b_2^*)^0 (+b_1 + a_2)^2 (+b_1^* + a_2^*)^0}_{a_1^l} \right| \cdot \left| \underbrace{(-a_1 + b_2)^1 (-a_1^* + b_2^*)^1 (-b_1 + a_2)^1 (-b_1^* + a_2^*)^1}_{a_1^r} \right| + \dots$$

(multiplied out and summed up) dimer ci vectors:

$$+2 \cdot \left| \underbrace{(a_1)^2 (b_2)^1 (b_1)^2 (a_2)^1 (a_1^*)^0 (b_2^*)^1 (b_1^*)^1 (a_2^*)^0}_{b_2} \right| + 2 \cdot \left| \underbrace{(a_1)^2 (b_2)^1 (b_1)^1 (a_2)^2 (a_1^*)^0 (b_2^*)^1 (b_1^*)^0 (a_2^*)^1}_{b_2} \right|$$

$$+2 \cdot \left| \underbrace{(a_1)^2 (b_2)^1 (b_1)^2 (a_2)^1 (a_1^*)^1 (b_2^*)^0 (b_1^*)^0 (a_2^*)^1}_{b_2} \right| + 2 \cdot \left| \underbrace{(a_1)^2 (b_2)^1 (b_1)^1 (a_2)^2 (a_1^*)^1 (b_2^*)^0 (b_1^*)^1 (a_2^*)^0}_{b_2} \right|$$

$$+2 \cdot \left| \underbrace{(a_1)^1 (b_2)^2 (b_1)^2 (a_2)^1 (a_1^*)^0 (b_2^*)^1 (b_1^*)^0 (a_2^*)^1}_{b_2} \right| + 2 \cdot \left| \underbrace{(a_1)^1 (b_2)^2 (b_1)^1 (a_2)^2 (a_1^*)^0 (b_2^*)^1 (b_1^*)^1 (a_2^*)^0}_{b_2} \right|$$

$$+2 \cdot \left| \underbrace{(a_1)^1 (b_2)^2 (b_1)^2 (a_2)^1 (a_1^*)^1 (b_2^*)^0 (b_1^*)^1 (a_2^*)^0}_{b_2} \right| + 2 \cdot \left| \underbrace{(a_1)^1 (b_2)^2 (b_1)^1 (a_2)^2 (a_1^*)^1 (b_2^*)^0 (b_1^*)^0 (a_2^*)^1}_{b_2} \right|$$

3.3 $i^3 a_1 * i^3 a_1$

$$i^3 a_1 * i^3 a_1$$

$$+ \left(\begin{array}{cccc} \uparrow & - & \uparrow & - \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{array} \right) - \left(\begin{array}{cccc} \uparrow & - & - & \uparrow \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{array} \right) - \left(\begin{array}{cccc} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{array} \right) + \left(\begin{array}{cccc} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{array} \right)$$

monomer ci vectors:

$$+ \left| \underbrace{(b_2^l)^1 (b_2^{*l})^1 (a_2^l)^2 (a_2^{*l})^0}_{a_1^l} \right| \cdot \left| \underbrace{(b_2^r)^1 (b_2^{*r})^1 (a_2^r)^2 (a_2^{*r})^0}_{a_1^r} \right| - \left| \underbrace{(b_2^l)^1 (b_2^{*l})^1 (a_2^l)^2 (a_2^{*l})^0}_{a_1^l} \right| \cdot \left| \underbrace{(b_2^r)^2 (b_2^{*r})^0 (a_2^r)^1 (a_2^{*r})^1}_{a_1^r} \right| \\ - \left| \underbrace{(b_2^l)^2 (b_2^{*l})^0 (a_2^l)^1 (a_2^{*l})^1}_{a_1^l} \right| \cdot \left| \underbrace{(b_2^r)^1 (b_2^{*r})^1 (a_2^r)^2 (a_2^{*r})^0}_{a_1^r} \right| + \left| \underbrace{(b_2^l)^2 (b_2^{*l})^0 (a_2^l)^1 (a_2^{*l})^1}_{a_1^l} \right| \cdot \left| \underbrace{(b_2^r)^2 (b_2^{*r})^0 (a_2^r)^1 (a_2^{*r})^1}_{a_1^r} \right|$$

substitution by dimer orbitals:

$$+ \left| \underbrace{(+a_1 + b_2)^1 (+a_1^* + b_2^*)^1 (+b_1 + a_2)^2 (+b_1^* + a_2^*)^0}_{a_1^l} \right| \cdot \left| \underbrace{(-a_1 + b_2)^1 (-a_1^* + b_2^*)^1 (-b_1 + a_2)^2 (-b_1^* + a_2^*)^0}_{a_1^r} \right| + \dots$$

(multiplied out and summed up) dimer ci vectors:

$$+ \left| \underbrace{(a_1)^1 (b_2)^1 (b_1)^2 (a_2)^2 (a_1^*)^1 (b_2^*)^1 (b_1^*)^0 (a_2^*)^0}_{a_1} \right| + 2 \cdot \left| \underbrace{(a_1)^1 (b_2)^2 (b_1)^2 (a_2)^1 (a_1^*)^1 (b_2^*)^0 (b_1^*)^0 (a_2^*)^1}_{a_1} \right| \\ + 2 \cdot \left| \underbrace{(a_1)^1 (b_2)^2 (b_1)^1 (a_2)^2 (a_1^*)^1 (b_2^*)^0 (b_1^*)^1 (a_2^*)^0}_{a_1} \right| - 2 \cdot \left| \underbrace{(a_1)^1 (b_2)^2 (b_1)^2 (a_2)^1 (a_1^*)^0 (b_2^*)^1 (b_1^*)^1 (a_2^*)^0}_{a_1} \right| \\ - 2 \cdot \left| \underbrace{(a_1)^1 (b_2)^2 (b_1)^1 (a_2)^2 (a_1^*)^0 (b_2^*)^1 (b_1^*)^0 (a_2^*)^1}_{a_1} \right| - 2 \cdot \left| \underbrace{(a_1)^2 (b_2)^1 (b_1)^2 (a_2)^1 (a_1^*)^1 (b_2^*)^0 (b_1^*)^1 (a_2^*)^0}_{a_1} \right| \\ - 2 \cdot \left| \underbrace{(a_1)^2 (b_2)^1 (b_1)^1 (a_2)^2 (a_1^*)^1 (b_2^*)^0 (b_1^*)^0 (a_2^*)^1}_{a_1} \right| + 2 \cdot \left| \underbrace{(a_1)^2 (b_2)^1 (b_1)^2 (a_2)^1 (a_1^*)^0 (b_2^*)^1 (b_1^*)^0 (a_2^*)^1}_{a_1} \right| \\ + 2 \cdot \left| \underbrace{(a_1)^2 (b_2)^1 (b_1)^1 (a_2)^2 (a_1^*)^0 (b_2^*)^1 (b_1^*)^1 (a_2^*)^0}_{a_1} \right| + \left| \underbrace{(a_1)^2 (b_2)^2 (b_1)^1 (a_2)^1 (a_1^*)^0 (b_2^*)^0 (b_1^*)^1 (a_2^*)^1}_{a_1} \right|$$

3.4 $i^3 a_1 * e^3 a_1 + e^3 a_1 * i^3 a_1$

$$i^3 a_1 * e^3 a_1 + e^3 a_1 * i^3 a_1$$

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow\uparrow & \uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow\uparrow & \uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow\uparrow & \uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \uparrow\uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} \\ - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \uparrow\uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow\uparrow & \uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \uparrow\uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \uparrow\uparrow & \uparrow & \uparrow\uparrow \end{pmatrix}$$

monomer ci vectors:

$$+ \left| \underbrace{(b_2^l)^1 (b_2^{*l})^1 (a_2^l)^2 (a_2^{*l})^0}_{a_1^l} \right| \cdot \left| \underbrace{(b_2^r)^1 (b_2^{*r})^1 (a_2^r)^2 (a_2^{*r})^0}_{a_1^r} \right| + \left| \underbrace{(b_2^l)^1 (b_2^{*l})^1 (a_2^l)^2 (a_2^{*l})^0}_{a_1^l} \right| \cdot \left| \underbrace{(b_2^r)^1 (b_2^{*r})^1 (a_2^r)^2 (a_2^{*r})^0}_{a_1^r} \right| \\ + \left| \underbrace{(b_2^l)^1 (b_2^{*l})^1 (a_2^l)^2 (a_2^{*l})^0}_{a_1^l} \right| \cdot \left| \underbrace{(b_2^r)^2 (b_2^{*r})^0 (a_2^r)^1 (a_2^{*r})^1}_{a_1^r} \right| + \left| \underbrace{(b_2^l)^2 (b_2^{*l})^0 (a_2^l)^1 (a_2^{*l})^1}_{a_1^l} \right| \cdot \left| \underbrace{(b_2^r)^1 (b_2^{*r})^1 (a_2^r)^2 (a_2^{*r})^0}_{a_1^r} \right| \\ - \left| \underbrace{(b_2^l)^2 (b_2^{*l})^0 (a_2^l)^1 (a_2^{*l})^1}_{a_1^l} \right| \cdot \left| \underbrace{(b_2^r)^1 (b_2^{*r})^1 (a_2^r)^2 (a_2^{*r})^0}_{a_1^r} \right| - \left| \underbrace{(b_2^l)^1 (b_2^{*l})^1 (a_2^l)^2 (a_2^{*l})^0}_{a_1^l} \right| \cdot \left| \underbrace{(b_2^r)^2 (b_2^{*r})^0 (a_2^r)^1 (a_2^{*r})^1}_{a_1^r} \right| \\ - \left| \underbrace{(b_2^l)^2 (b_2^{*l})^0 (a_2^l)^1 (a_2^{*l})^1}_{a_1^l} \right| \cdot \left| \underbrace{(b_2^r)^2 (b_2^{*r})^0 (a_2^r)^1 (a_2^{*r})^1}_{a_1^r} \right| - \left| \underbrace{(b_2^l)^2 (b_2^{*l})^0 (a_2^l)^1 (a_2^{*l})^1}_{a_1^l} \right| \cdot \left| \underbrace{(b_2^r)^2 (b_2^{*r})^0 (a_2^r)^1 (a_2^{*r})^1}_{a_1^r} \right|$$

substitution by dimer orbitals:

$$+ \left| \underbrace{(+a_1 + b_2)^1 (+a_1^* + b_2^*)^1 (+b_1 + a_2)^2 (+b_1^* + a_2^*)^0}_{a_1^l} \right| \cdot \left| \underbrace{(-a_1 + b_2)^1 (-a_1^* + b_2^*)^1 (-b_1 + a_2)^2 (-b_1^* + a_2^*)^0}_{a_1^r} \right| + \dots$$

(multiplied out and summed up) dimer ci vectors:

$$+ 2 \cdot \left| \underbrace{(a_1)^1 (b_2)^1 (b_1)^2 (a_2)^2 (a_1^*)^1 (b_2^*)^1 (b_1^*)^0 (a_2^*)^0}_{a_1} \right| - 2 \cdot \left| \underbrace{(a_1)^2 (b_2)^2 (b_1)^1 (a_2)^1 (a_1^*)^0 (b_2^*)^0 (b_1^*)^1 (a_2^*)^1}_{a_1} \right|$$

3.5 $i^3 a_1 * e^3 a_1 - e^3 a_1 * i^3 a_1$

$$i^3 a_1 * e^3 a_1 - e^3 a_1 * i^3 a_1$$

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow\uparrow & \uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow\uparrow & \uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow\uparrow & \uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \uparrow\uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} \\ - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \uparrow\uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow\uparrow & \uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \uparrow\uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \uparrow\uparrow & \uparrow & \uparrow\uparrow \end{pmatrix}$$

monomer ci vectors:

$$+ \left| \underbrace{(b_2^l)^1 (b_2^{*l})^1 (a_2^l)^2 (a_2^{*l})^0}_{a_1^l} \right| \cdot \left| \underbrace{(b_2^r)^1 (b_2^{*r})^1 (a_2^r)^2 (a_2^{*r})^0}_{a_1^r} \right| - \left| \underbrace{(b_2^l)^1 (b_2^{*l})^1 (a_2^l)^2 (a_2^{*l})^0}_{a_1^l} \right| \cdot \left| \underbrace{(b_2^r)^1 (b_2^{*r})^1 (a_2^r)^2 (a_2^{*r})^0}_{a_1^r} \right| \\ + \left| \underbrace{(b_2^l)^1 (b_2^{*l})^1 (a_2^l)^2 (a_2^{*l})^0}_{a_1^l} \right| \cdot \left| \underbrace{(b_2^r)^2 (b_2^{*r})^0 (a_2^r)^1 (a_2^{*r})^1}_{a_1^r} \right| - \left| \underbrace{(b_2^l)^2 (b_2^{*l})^0 (a_2^l)^1 (a_2^{*l})^1}_{a_1^l} \right| \cdot \left| \underbrace{(b_2^r)^1 (b_2^{*r})^1 (a_2^r)^2 (a_2^{*r})^0}_{a_1^r} \right| \\ - \left| \underbrace{(b_2^l)^2 (b_2^{*l})^0 (a_2^l)^1 (a_2^{*l})^1}_{a_1^l} \right| \cdot \left| \underbrace{(b_2^r)^1 (b_2^{*r})^1 (a_2^r)^2 (a_2^{*r})^0}_{a_1^r} \right| + \left| \underbrace{(b_2^l)^1 (b_2^{*l})^1 (a_2^l)^2 (a_2^{*l})^0}_{a_1^l} \right| \cdot \left| \underbrace{(b_2^r)^2 (b_2^{*r})^0 (a_2^r)^1 (a_2^{*r})^1}_{a_1^r} \right| \\ - \left| \underbrace{(b_2^l)^2 (b_2^{*l})^0 (a_2^l)^1 (a_2^{*l})^1}_{a_1^l} \right| \cdot \left| \underbrace{(b_2^r)^2 (b_2^{*r})^0 (a_2^r)^1 (a_2^{*r})^1}_{a_1^r} \right| + \left| \underbrace{(b_2^l)^2 (b_2^{*l})^0 (a_2^l)^1 (a_2^{*l})^1}_{a_1^l} \right| \cdot \left| \underbrace{(b_2^r)^2 (b_2^{*r})^0 (a_2^r)^1 (a_2^{*r})^1}_{a_1^r} \right|$$

substitution by dimer orbitals:

$$+ \left| \underbrace{(+a_1 + b_2)^1 (+a_1^* + b_2^*)^1 (+b_1 + a_2)^2 (+b_1^* + a_2^*)^0}_{a_1^l} \right| \cdot \left| \underbrace{(-a_1 + b_2)^1 (-a_1^* + b_2^*)^1 (-b_1 + a_2)^2 (-b_1^* + a_2^*)^0}_{a_1^r} \right| + \dots$$

(multiplied out and summed up) dimer ci vectors:

$$+4 \cdot \left| \underbrace{(a_1)^1 (b_2)^2 (b_1)^2 (a_2)^1 (a_1^*)^1 (b_2^*)^0 (b_1^*)^1 (a_2^*)^0}_{b_2} \right| + 4 \cdot \left| \underbrace{(a_1)^1 (b_2)^2 (b_1)^1 (a_2)^2 (a_1^*)^1 (b_2^*)^0 (b_1^*)^0 (a_2^*)^1}_{b_2} \right| \\ - 4 \cdot \left| \underbrace{(a_1)^1 (b_2)^2 (b_1)^2 (a_2)^1 (a_1^*)^0 (b_2^*)^1 (b_1^*)^0 (a_2^*)^1}_{b_2} \right| - 4 \cdot \left| \underbrace{(a_1)^1 (b_2)^2 (b_1)^1 (a_2)^2 (a_1^*)^0 (b_2^*)^1 (b_1^*)^1 (a_2^*)^0}_{b_2} \right| \\ - 4 \cdot \left| \underbrace{(a_1)^2 (b_2)^1 (b_1)^2 (a_2)^1 (a_1^*)^1 (b_2^*)^0 (b_1^*)^0 (a_2^*)^1}_{b_2} \right| - 4 \cdot \left| \underbrace{(a_1)^2 (b_2)^1 (b_1)^1 (a_2)^2 (a_1^*)^1 (b_2^*)^0 (b_1^*)^1 (a_2^*)^0}_{b_2} \right| \\ + 4 \cdot \left| \underbrace{(a_1)^2 (b_2)^1 (b_1)^2 (a_2)^1 (a_1^*)^0 (b_2^*)^1 (b_1^*)^1 (a_2^*)^0}_{b_2} \right| + 4 \cdot \left| \underbrace{(a_1)^2 (b_2)^1 (b_1)^1 (a_2)^2 (a_1^*)^0 (b_2^*)^1 (b_1^*)^0 (a_2^*)^1}_{b_2} \right|$$

3.6 $i^3 a_1 * i^3 b_1 + i^3 b_1 * i^3 a_1$

$$i^3 a_1 * i^3 b_1 + i^3 b_1 * i^3 a_1$$

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} \\ - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix}$$

monomer ci vectors:

$$+ \left| \underbrace{(b_2^l)^1 (b_2^{*l})^1 (a_2^l)^2 (a_2^{*l})^0}_{a_1^l} \cdot \underbrace{(b_2^r)^2 (b_2^{*r})^1 (a_2^r)^1 (a_2^{*r})^0}_{b_1^r} \right| + \left| \underbrace{(b_2^l)^2 (b_2^{*l})^1 (a_2^l)^1 (a_2^{*l})^0}_{b_1^l} \cdot \underbrace{(b_2^r)^1 (b_2^{*r})^1 (a_2^r)^2 (a_2^{*r})^0}_{a_1^r} \right| \\ - \left| \underbrace{(b_2^l)^1 (b_2^{*l})^1 (a_2^l)^2 (a_2^{*l})^0}_{a_1^l} \cdot \underbrace{(b_2^r)^1 (b_2^{*r})^0 (a_2^r)^2 (a_2^{*r})^1}_{b_1^r} \right| - \left| \underbrace{(b_2^l)^1 (b_2^{*l})^0 (a_2^l)^2 (a_2^{*l})^1}_{b_1^l} \cdot \underbrace{(b_2^r)^1 (b_2^{*r})^1 (a_2^r)^2 (a_2^{*r})^0}_{a_1^r} \right| \\ - \left| \underbrace{(b_2^l)^2 (b_2^{*l})^0 (a_2^l)^1 (a_2^{*l})^1}_{a_1^l} \cdot \underbrace{(b_2^r)^2 (b_2^{*r})^1 (a_2^r)^1 (a_2^{*r})^0}_{b_1^r} \right| - \left| \underbrace{(b_2^l)^2 (b_2^{*l})^1 (a_2^l)^1 (a_2^{*l})^0}_{b_1^l} \cdot \underbrace{(b_2^r)^2 (b_2^{*r})^0 (a_2^r)^1 (a_2^{*r})^1}_{a_1^r} \right| \\ + \left| \underbrace{(b_2^l)^2 (b_2^{*l})^0 (a_2^l)^1 (a_2^{*l})^1}_{a_1^l} \cdot \underbrace{(b_2^r)^1 (b_2^{*r})^0 (a_2^r)^2 (a_2^{*r})^1}_{b_1^r} \right| + \left| \underbrace{(b_2^l)^1 (b_2^{*l})^0 (a_2^l)^2 (a_2^{*l})^1}_{b_1^l} \cdot \underbrace{(b_2^r)^2 (b_2^{*r})^0 (a_2^r)^1 (a_2^{*r})^1}_{a_1^r} \right|$$

substitution by dimer orbitals:

$$+ \left| \underbrace{(+a_1 + b_2)^1 (+a_1^* + b_2^*)^1 (+b_1 + a_2)^2 (+b_1^* + a_2^*)^0}_{a_1^l} \right| \cdot \left| \underbrace{(-a_1 + b_2)^2 (-a_1^* + b_2^*)^1 (-b_1 + a_2)^1 (-b_1^* + a_2^*)^0}_{b_1^r} \right| + \dots$$

(multiplied out and summed up) dimer ci vectors:

$$+2 \cdot \left| \underbrace{(a_1)^1 (b_2)^2 (b_1)^2 (a_2)^1 (a_1^*)^1 (b_2^*)^1 (b_1^*)^0 (a_2^*)^0}_{b_1} \right| - 2 \cdot \left| \underbrace{(a_1)^2 (b_2)^1 (b_1)^1 (a_2)^2 (a_1^*)^1 (b_2^*)^1 (b_1^*)^0 (a_2^*)^0}_{b_1} \right| \\ - 2 \cdot \left| \underbrace{(a_1)^1 (b_2)^1 (b_1)^2 (a_2)^2 (a_1^*)^1 (b_2^*)^0 (b_1^*)^0 (a_2^*)^1}_{b_1} \right| + 2 \cdot \left| \underbrace{(a_1)^1 (b_2)^1 (b_1)^2 (a_2)^2 (a_1^*)^0 (b_2^*)^1 (b_1^*)^1 (a_2^*)^0}_{b_1} \right| \\ + 2 \cdot \left| \underbrace{(a_1)^2 (b_2)^2 (b_1)^1 (a_2)^1 (a_1^*)^0 (b_2^*)^1 (b_1^*)^1 (a_2^*)^0}_{b_1} \right| - 2 \cdot \left| \underbrace{(a_1)^2 (b_2)^2 (b_1)^1 (a_2)^1 (a_1^*)^1 (b_2^*)^0 (b_1^*)^0 (a_2^*)^1}_{b_1} \right| \\ - 2 \cdot \left| \underbrace{(a_1)^2 (b_2)^1 (b_1)^1 (a_2)^2 (a_1^*)^0 (b_2^*)^0 (b_1^*)^1 (a_2^*)^1}_{b_1} \right| + 2 \cdot \left| \underbrace{(a_1)^1 (b_2)^2 (b_1)^2 (a_2)^1 (a_1^*)^0 (b_2^*)^0 (b_1^*)^1 (a_2^*)^1}_{b_1} \right|$$

3.7 $i^3 a_1 * i^3 b_1 - i^3 b_1 * i^3 a_1$

$$i^3 a_1 * i^3 b_1 - i^3 b_1 * i^3 a_1$$

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} \\ - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix}$$

monomer ci vectors:

$$+ \left| \underbrace{(b_2^l)^1 (b_2^{*l})^1 (a_2^l)^2 (a_2^{*l})^0}_{a_1^l} \cdot \underbrace{(b_2^r)^2 (b_2^{*r})^1 (a_2^r)^1 (a_2^{*r})^0}_{b_1^r} \right| - \left| \underbrace{(b_2^l)^2 (b_2^{*l})^1 (a_2^l)^1 (a_2^{*l})^0}_{b_1^l} \cdot \underbrace{(b_2^r)^1 (b_2^{*r})^1 (a_2^r)^2 (a_2^{*r})^0}_{a_1^r} \right| \\ - \left| \underbrace{(b_2^l)^1 (b_2^{*l})^1 (a_2^l)^2 (a_2^{*l})^0}_{a_1^l} \cdot \underbrace{(b_2^r)^1 (b_2^{*r})^0 (a_2^r)^2 (a_2^{*r})^1}_{b_1^r} \right| + \left| \underbrace{(b_2^l)^1 (b_2^{*l})^0 (a_2^l)^2 (a_2^{*l})^1}_{b_1^l} \cdot \underbrace{(b_2^r)^1 (b_2^{*r})^1 (a_2^r)^2 (a_2^{*r})^0}_{a_1^r} \right| \\ - \left| \underbrace{(b_2^l)^2 (b_2^{*l})^0 (a_2^l)^1 (a_2^{*l})^1}_{a_1^l} \cdot \underbrace{(b_2^r)^2 (b_2^{*r})^1 (a_2^r)^1 (a_2^{*r})^0}_{b_1^r} \right| + \left| \underbrace{(b_2^l)^2 (b_2^{*l})^1 (a_2^l)^1 (a_2^{*l})^0}_{b_1^l} \cdot \underbrace{(b_2^r)^2 (b_2^{*r})^0 (a_2^r)^1 (a_2^{*r})^1}_{a_1^r} \right| \\ + \left| \underbrace{(b_2^l)^2 (b_2^{*l})^0 (a_2^l)^1 (a_2^{*l})^1}_{a_1^l} \cdot \underbrace{(b_2^r)^1 (b_2^{*r})^0 (a_2^r)^2 (a_2^{*r})^1}_{b_1^r} \right| - \left| \underbrace{(b_2^l)^1 (b_2^{*l})^0 (a_2^l)^2 (a_2^{*l})^1}_{b_1^l} \cdot \underbrace{(b_2^r)^2 (b_2^{*r})^0 (a_2^r)^1 (a_2^{*r})^1}_{a_1^r} \right|$$

substitution by dimer orbitals:

$$+ \left| \underbrace{(+a_1 + b_2)^1 (+a_1^* + b_2^*)^1 (+b_1 + a_2)^2 (+b_1^* + a_2^*)^0}_{a_1^l} \right| \cdot \left| \underbrace{(-a_1 + b_2)^2 (-a_1^* + b_2^*)^1 (-b_1 + a_2)^1 (-b_1^* + a_2^*)^0}_{b_1^r} \right| + \dots$$

(multiplied out and summed up) dimer ci vectors:

$$-2 \cdot \left| \underbrace{(a_1)^1 (b_2)^2 (b_1)^1 (a_2)^2 (a_1^*)^1 (b_2^*)^1 (b_1^*)^0 (a_2^*)^0}_{a_2} \right| + 2 \cdot \left| \underbrace{(a_1)^2 (b_2)^1 (b_1)^2 (a_2)^1 (a_1^*)^1 (b_2^*)^1 (b_1^*)^0 (a_2^*)^0}_{a_2} \right| \\ + 2 \cdot \left| \underbrace{(a_1)^1 (b_2)^1 (b_1)^2 (a_2)^2 (a_1^*)^1 (b_2^*)^0 (b_1^*)^1 (a_2^*)^0}_{a_2} \right| - 2 \cdot \left| \underbrace{(a_1)^1 (b_2)^1 (b_1)^2 (a_2)^2 (a_1^*)^0 (b_2^*)^1 (b_1^*)^0 (a_2^*)^1}_{a_2} \right| \\ + 2 \cdot \left| \underbrace{(a_1)^2 (b_2)^2 (b_1)^1 (a_2)^1 (a_1^*)^0 (b_2^*)^1 (b_1^*)^0 (a_2^*)^1}_{a_2} \right| - 2 \cdot \left| \underbrace{(a_1)^2 (b_2)^2 (b_1)^1 (a_2)^1 (a_1^*)^1 (b_2^*)^0 (b_1^*)^1 (a_2^*)^0}_{a_2} \right| \\ - 2 \cdot \left| \underbrace{(a_1)^2 (b_2)^1 (b_1)^2 (a_2)^1 (a_1^*)^0 (b_2^*)^0 (b_1^*)^1 (a_2^*)^1}_{a_2} \right| + 2 \cdot \left| \underbrace{(a_1)^1 (b_2)^2 (b_1)^1 (a_2)^2 (a_1^*)^0 (b_2^*)^0 (b_1^*)^1 (a_2^*)^1}_{a_2} \right|$$

3.8 $i^3 a_1 * e^3 b_1 + e^3 b_1 * i^3 a_1$

$$i^3 a_1 * e^3 b_1 + e^3 b_1 * i^3 a_1$$

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} \\ - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix}$$

monomer ci vectors:

$$+ \left| \underbrace{(b_2^l)^1 (b_2^{*l})^1 (a_2^l)^2 (a_2^{*l})^0}_{a_1^l} \cdot \underbrace{(b_2^r)^2 (b_2^{*r})^1 (a_2^r)^1 (a_2^{*r})^0}_{b_1^r} \right| + \left| \underbrace{(b_2^l)^2 (b_2^{*l})^1 (a_2^l)^1 (a_2^{*l})^0}_{b_1^l} \cdot \underbrace{(b_2^r)^1 (b_2^{*r})^1 (a_2^r)^2 (a_2^{*r})^0}_{a_1^r} \right| \\ + \left| \underbrace{(b_2^l)^1 (b_2^{*l})^1 (a_2^l)^2 (a_2^{*l})^0}_{a_1^l} \cdot \underbrace{(b_2^r)^1 (b_2^{*r})^0 (a_2^r)^2 (a_2^{*r})^1}_{b_1^r} \right| + \left| \underbrace{(b_2^l)^1 (b_2^{*l})^0 (a_2^l)^2 (a_2^{*l})^1}_{b_1^l} \cdot \underbrace{(b_2^r)^1 (b_2^{*r})^1 (a_2^r)^2 (a_2^{*r})^0}_{a_1^r} \right| \\ - \left| \underbrace{(b_2^l)^2 (b_2^{*l})^0 (a_2^l)^1 (a_2^{*l})^1}_{a_1^l} \cdot \underbrace{(b_2^r)^2 (b_2^{*r})^1 (a_2^r)^1 (a_2^{*r})^0}_{b_1^r} \right| - \left| \underbrace{(b_2^l)^2 (b_2^{*l})^1 (a_2^l)^1 (a_2^{*l})^0}_{b_1^l} \cdot \underbrace{(b_2^r)^2 (b_2^{*r})^0 (a_2^r)^1 (a_2^{*r})^1}_{a_1^r} \right| \\ - \left| \underbrace{(b_2^l)^2 (b_2^{*l})^0 (a_2^l)^1 (a_2^{*l})^1}_{a_1^l} \cdot \underbrace{(b_2^r)^1 (b_2^{*r})^0 (a_2^r)^2 (a_2^{*r})^1}_{b_1^r} \right| - \left| \underbrace{(b_2^l)^1 (b_2^{*l})^0 (a_2^l)^2 (a_2^{*l})^1}_{b_1^l} \cdot \underbrace{(b_2^r)^2 (b_2^{*r})^0 (a_2^r)^1 (a_2^{*r})^1}_{a_1^r} \right|$$

substitution by dimer orbitals:

$$+ \left| \underbrace{(+a_1 + b_2)^1 (+a_1^* + b_2^*)^1 (+b_1 + a_2)^2 (+b_1^* + a_2^*)^0}_{a_1^l} \right| \cdot \left| \underbrace{(-a_1 + b_2)^2 (-a_1^* + b_2^*)^1 (-b_1 + a_2)^1 (-b_1^* + a_2^*)^0}_{b_1^r} \right| + \dots$$

(multiplied out and summed up) dimer ci vectors:

$$+2 \cdot \left| \underbrace{(a_1)^1 (b_2)^2 (b_1)^2 (a_2)^1 (a_1^*)^1 (b_2^*)^1 (b_1^*)^0 (a_2^*)^0}_{b_1} \right| - 2 \cdot \left| \underbrace{(a_1)^2 (b_2)^1 (b_1)^1 (a_2)^2 (a_1^*)^1 (b_2^*)^1 (b_1^*)^0 (a_2^*)^0}_{b_1} \right| \\ +2 \cdot \left| \underbrace{(a_1)^1 (b_2)^1 (b_1)^2 (a_2)^2 (a_1^*)^1 (b_2^*)^0 (b_1^*)^0 (a_2^*)^1}_{b_1} \right| - 2 \cdot \left| \underbrace{(a_1)^1 (b_2)^1 (b_1)^2 (a_2)^2 (a_1^*)^0 (b_2^*)^1 (b_1^*)^1 (a_2^*)^0}_{b_1} \right| \\ +2 \cdot \left| \underbrace{(a_1)^2 (b_2)^2 (b_1)^1 (a_2)^1 (a_1^*)^0 (b_2^*)^1 (b_1^*)^1 (a_2^*)^0}_{b_1} \right| - 2 \cdot \left| \underbrace{(a_1)^2 (b_2)^2 (b_1)^1 (a_2)^1 (a_1^*)^1 (b_2^*)^0 (b_1^*)^0 (a_2^*)^1}_{b_1} \right| \\ +2 \cdot \left| \underbrace{(a_1)^2 (b_2)^1 (b_1)^1 (a_2)^2 (a_1^*)^0 (b_2^*)^0 (b_1^*)^1 (a_2^*)^1}_{b_1} \right| - 2 \cdot \left| \underbrace{(a_1)^1 (b_2)^2 (b_1)^2 (a_2)^1 (a_1^*)^0 (b_2^*)^0 (b_1^*)^1 (a_2^*)^1}_{b_1} \right|$$

3.9 $i^3 a_1 * e^3 b_1 - e^3 b_1 * i^3 a_1$

$$i^3 a_1 * e^3 b_1 - e^3 b_1 * i^3 a_1$$

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} \\ - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix}$$

monomer ci vectors:

$$+ \left| \underbrace{(b_2^l)^1 (b_2^{*l})^1 (a_2^l)^2 (a_2^{*l})^0}_{a_1^l} \cdot \underbrace{(b_2^r)^2 (b_2^{*r})^1 (a_2^r)^1 (a_2^{*r})^0}_{b_1^r} \right| - \left| \underbrace{(b_2^l)^2 (b_2^{*l})^1 (a_2^l)^1 (a_2^{*l})^0}_{b_1^l} \cdot \underbrace{(b_2^r)^1 (b_2^{*r})^1 (a_2^r)^2 (a_2^{*r})^0}_{a_1^r} \right| \\ + \left| \underbrace{(b_2^l)^1 (b_2^{*l})^1 (a_2^l)^2 (a_2^{*l})^0}_{a_1^l} \cdot \underbrace{(b_2^r)^1 (b_2^{*r})^0 (a_2^r)^2 (a_2^{*r})^1}_{b_1^r} \right| - \left| \underbrace{(b_2^l)^1 (b_2^{*l})^0 (a_2^l)^2 (a_2^{*l})^1}_{b_1^l} \cdot \underbrace{(b_2^r)^1 (b_2^{*r})^1 (a_2^r)^2 (a_2^{*r})^0}_{a_1^r} \right| \\ - \left| \underbrace{(b_2^l)^2 (b_2^{*l})^0 (a_2^l)^1 (a_2^{*l})^1}_{a_1^l} \cdot \underbrace{(b_2^r)^2 (b_2^{*r})^1 (a_2^r)^1 (a_2^{*r})^0}_{b_1^r} \right| + \left| \underbrace{(b_2^l)^2 (b_2^{*l})^1 (a_2^l)^1 (a_2^{*l})^0}_{b_1^l} \cdot \underbrace{(b_2^r)^2 (b_2^{*r})^0 (a_2^r)^1 (a_2^{*r})^1}_{a_1^r} \right| \\ - \left| \underbrace{(b_2^l)^2 (b_2^{*l})^0 (a_2^l)^1 (a_2^{*l})^1}_{a_1^l} \cdot \underbrace{(b_2^r)^1 (b_2^{*r})^0 (a_2^r)^2 (a_2^{*r})^1}_{b_1^r} \right| + \left| \underbrace{(b_2^l)^1 (b_2^{*l})^0 (a_2^l)^2 (a_2^{*l})^1}_{b_1^l} \cdot \underbrace{(b_2^r)^2 (b_2^{*r})^0 (a_2^r)^1 (a_2^{*r})^1}_{a_1^r} \right|$$

substitution by dimer orbitals:

$$+ \left| \underbrace{(+a_1 + b_2)^1 (+a_1^* + b_2^*)^1 (+b_1 + a_2)^2 (+b_1^* + a_2^*)^0}_{a_1^l} \right| \cdot \left| \underbrace{(-a_1 + b_2)^2 (-a_1^* + b_2^*)^1 (-b_1 + a_2)^1 (-b_1^* + a_2^*)^0}_{b_1^r} \right| + \dots$$

(multiplied out and summed up) dimer ci vectors:

$$-2 \cdot \left| \underbrace{(a_1)^1 (b_2)^2 (b_1)^1 (a_2)^2 (a_1^*)^1 (b_2^*)^1 (b_1^*)^0 (a_2^*)^0}_{a_2} \right| + 2 \cdot \left| \underbrace{(a_1)^2 (b_2)^1 (b_1)^2 (a_2)^1 (a_1^*)^1 (b_2^*)^1 (b_1^*)^0 (a_2^*)^0}_{a_2} \right| \\ -2 \cdot \left| \underbrace{(a_1)^1 (b_2)^1 (b_1)^2 (a_2)^2 (a_1^*)^1 (b_2^*)^0 (b_1^*)^1 (a_2^*)^0}_{a_2} \right| + 2 \cdot \left| \underbrace{(a_1)^1 (b_2)^1 (b_1)^2 (a_2)^2 (a_1^*)^0 (b_2^*)^1 (b_1^*)^0 (a_2^*)^1}_{a_2} \right| \\ +2 \cdot \left| \underbrace{(a_1)^2 (b_2)^2 (b_1)^1 (a_2)^1 (a_1^*)^0 (b_2^*)^1 (b_1^*)^0 (a_2^*)^1}_{a_2} \right| - 2 \cdot \left| \underbrace{(a_1)^2 (b_2)^2 (b_1)^1 (a_2)^1 (a_1^*)^1 (b_2^*)^0 (b_1^*)^1 (a_2^*)^0}_{a_2} \right| \\ +2 \cdot \left| \underbrace{(a_1)^2 (b_2)^1 (b_1)^2 (a_2)^1 (a_1^*)^0 (b_2^*)^0 (b_1^*)^1 (a_2^*)^1}_{a_2} \right| - 2 \cdot \left| \underbrace{(a_1)^1 (b_2)^2 (b_1)^1 (a_2)^2 (a_1^*)^0 (b_2^*)^0 (b_1^*)^1 (a_2^*)^1}_{a_2} \right|$$

3.10 $e^3 a_1 * e^3 a_1$

$$e^3 a_1 * e^3 a_1$$

$$+ \left(\begin{array}{cccc} \uparrow & - & \uparrow & - \\ \uparrow\uparrow & \uparrow & \uparrow\uparrow & \uparrow \end{array} \right) + \left(\begin{array}{cccc} \uparrow & - & - & \uparrow \\ \uparrow\uparrow & \uparrow & \uparrow & \uparrow\uparrow \end{array} \right) + \left(\begin{array}{cccc} - & \uparrow & \uparrow & - \\ \uparrow & \uparrow\uparrow & \uparrow\uparrow & \uparrow \end{array} \right) + \left(\begin{array}{cccc} - & \uparrow & - & \uparrow \\ \uparrow & \uparrow\uparrow & \uparrow & \uparrow\uparrow \end{array} \right)$$

monomer ci vectors:

$$+ \left| \underbrace{(b_2^l)^1 (b_2^{*l})^1 (a_2^l)^2 (a_2^{*l})^0}_{a_1^l} \right| \cdot \left| \underbrace{(b_2^r)^1 (b_2^{*r})^1 (a_2^r)^2 (a_2^{*r})^0}_{a_1^r} \right| + \left| \underbrace{(b_2^l)^1 (b_2^{*l})^1 (a_2^l)^2 (a_2^{*l})^0}_{a_1^l} \right| \cdot \left| \underbrace{(b_2^r)^2 (b_2^{*r})^0 (a_2^r)^1 (a_2^{*r})^1}_{a_1^r} \right|$$

$$+ \left| \underbrace{(b_2^l)^2 (b_2^{*l})^0 (a_2^l)^1 (a_2^{*l})^1}_{a_1^l} \right| \cdot \left| \underbrace{(b_2^r)^1 (b_2^{*r})^1 (a_2^r)^2 (a_2^{*r})^0}_{a_1^r} \right| + \left| \underbrace{(b_2^l)^2 (b_2^{*l})^0 (a_2^l)^1 (a_2^{*l})^1}_{a_1^l} \right| \cdot \left| \underbrace{(b_2^r)^2 (b_2^{*r})^0 (a_2^r)^1 (a_2^{*r})^1}_{a_1^r} \right|$$

substitution by dimer orbitals:

$$+ \left| \underbrace{(+a_1 + b_2)^1 (+a_1^* + b_2^*)^1 (+b_1 + a_2)^2 (+b_1^* + a_2^*)^0}_{a_1^l} \right| \cdot \left| \underbrace{(-a_1 + b_2)^1 (-a_1^* + b_2^*)^1 (-b_1 + a_2)^2 (-b_1^* + a_2^*)^0}_{a_1^r} \right| + \dots$$

(multiplied out and summed up) dimer ci vectors:

$$+ \left| \underbrace{(a_1)^1 (b_2)^1 (b_1)^2 (a_2)^2 (a_1^*)^1 (b_2^*)^1 (b_1^*)^0 (a_2^*)^0}_{a_1} \right| - 2 \cdot \left| \underbrace{(a_1)^1 (b_2)^2 (b_1)^2 (a_2)^1 (a_1^*)^1 (b_2^*)^0 (b_1^*)^0 (a_2^*)^1}_{a_1} \right|$$

$$- 2 \cdot \left| \underbrace{(a_1)^1 (b_2)^2 (b_1)^1 (a_2)^2 (a_1^*)^1 (b_2^*)^0 (b_1^*)^1 (a_2^*)^0}_{a_1} \right| + 2 \cdot \left| \underbrace{(a_1)^1 (b_2)^2 (b_1)^2 (a_2)^1 (a_1^*)^0 (b_2^*)^1 (b_1^*)^1 (a_2^*)^0}_{a_1} \right|$$

$$+ 2 \cdot \left| \underbrace{(a_1)^1 (b_2)^2 (b_1)^1 (a_2)^2 (a_1^*)^0 (b_2^*)^1 (b_1^*)^0 (a_2^*)^1}_{a_1} \right| + 2 \cdot \left| \underbrace{(a_1)^2 (b_2)^1 (b_1)^2 (a_2)^1 (a_1^*)^1 (b_2^*)^0 (b_1^*)^1 (a_2^*)^0}_{a_1} \right|$$

$$+ 2 \cdot \left| \underbrace{(a_1)^2 (b_2)^1 (b_1)^1 (a_2)^2 (a_1^*)^1 (b_2^*)^0 (b_1^*)^0 (a_2^*)^1}_{a_1} \right| - 2 \cdot \left| \underbrace{(a_1)^2 (b_2)^1 (b_1)^2 (a_2)^1 (a_1^*)^0 (b_2^*)^1 (b_1^*)^0 (a_2^*)^1}_{a_1} \right|$$

$$- 2 \cdot \left| \underbrace{(a_1)^2 (b_2)^1 (b_1)^1 (a_2)^2 (a_1^*)^0 (b_2^*)^1 (b_1^*)^1 (a_2^*)^0}_{a_1} \right| + \left| \underbrace{(a_1)^2 (b_2)^2 (b_1)^1 (a_2)^1 (a_1^*)^0 (b_2^*)^0 (b_1^*)^1 (a_2^*)^1}_{a_1} \right|$$

3.11 $e^3 a_1 * i^3 b_1 + i^3 b_1 * e^3 a_1$

$$e^3 a_1 * i^3 b_1 + i^3 b_1 * e^3 a_1$$

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} \\ + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix}$$

monomer ci vectors:

$$+ \left| \underbrace{(b_2^l)^1 (b_2^{*l})^1 (a_2^l)^2 (a_2^{*l})^0}_{a_1^l} \right| \cdot \left| \underbrace{(b_2^r)^2 (b_2^{*r})^1 (a_2^r)^1 (a_2^{*r})^0}_{b_1^r} \right| + \left| \underbrace{(b_2^l)^2 (b_2^{*l})^1 (a_2^l)^1 (a_2^{*l})^0}_{b_1^l} \right| \cdot \left| \underbrace{(b_2^r)^1 (b_2^{*r})^1 (a_2^r)^2 (a_2^{*r})^0}_{a_1^r} \right| \\ - \left| \underbrace{(b_2^l)^1 (b_2^{*l})^1 (a_2^l)^2 (a_2^{*l})^0}_{a_1^l} \right| \cdot \left| \underbrace{(b_2^r)^1 (b_2^{*r})^0 (a_2^r)^2 (a_2^{*r})^1}_{b_1^r} \right| - \left| \underbrace{(b_2^l)^1 (b_2^{*l})^0 (a_2^l)^2 (a_2^{*l})^1}_{b_1^l} \right| \cdot \left| \underbrace{(b_2^r)^1 (b_2^{*r})^1 (a_2^r)^2 (a_2^{*r})^0}_{a_1^r} \right| \\ + \left| \underbrace{(b_2^l)^2 (b_2^{*l})^0 (a_2^l)^1 (a_2^{*l})^1}_{a_1^l} \right| \cdot \left| \underbrace{(b_2^r)^2 (b_2^{*r})^1 (a_2^r)^1 (a_2^{*r})^0}_{b_1^r} \right| + \left| \underbrace{(b_2^l)^2 (b_2^{*l})^1 (a_2^l)^1 (a_2^{*l})^0}_{b_1^l} \right| \cdot \left| \underbrace{(b_2^r)^2 (b_2^{*r})^0 (a_2^r)^1 (a_2^{*r})^1}_{a_1^r} \right| \\ - \left| \underbrace{(b_2^l)^2 (b_2^{*l})^0 (a_2^l)^1 (a_2^{*l})^1}_{a_1^l} \right| \cdot \left| \underbrace{(b_2^r)^1 (b_2^{*r})^0 (a_2^r)^2 (a_2^{*r})^1}_{b_1^r} \right| - \left| \underbrace{(b_2^l)^1 (b_2^{*l})^0 (a_2^l)^2 (a_2^{*l})^1}_{b_1^l} \right| \cdot \left| \underbrace{(b_2^r)^2 (b_2^{*r})^0 (a_2^r)^1 (a_2^{*r})^1}_{a_1^r} \right|$$

substitution by dimer orbitals:

$$+ \left| \underbrace{(+a_1 + b_2)^1 (+a_1^* + b_2^*)^1 (+b_1 + a_2)^2 (+b_1^* + a_2^*)^0}_{a_1^l} \right| \cdot \left| \underbrace{(-a_1 + b_2)^2 (-a_1^* + b_2^*)^1 (-b_1 + a_2)^1 (-b_1^* + a_2^*)^0}_{b_1^r} \right| + \dots$$

(multiplied out and summed up) dimer ci vectors:

$$+2 \cdot \left| \underbrace{(a_1)^1 (b_2)^2 (b_1)^2 (a_2)^1 (a_1^*)^1 (b_2^*)^1 (b_1^*)^0 (a_2^*)^0}_{b_1} \right| - 2 \cdot \left| \underbrace{(a_1)^2 (b_2)^1 (b_1)^1 (a_2)^2 (a_1^*)^1 (b_2^*)^1 (b_1^*)^0 (a_2^*)^0}_{b_1} \right| \\ - 2 \cdot \left| \underbrace{(a_1)^1 (b_2)^1 (b_1)^2 (a_2)^2 (a_1^*)^1 (b_2^*)^0 (b_1^*)^0 (a_2^*)^1}_{b_1} \right| + 2 \cdot \left| \underbrace{(a_1)^1 (b_2)^1 (b_1)^2 (a_2)^2 (a_1^*)^0 (b_2^*)^1 (b_1^*)^1 (a_2^*)^0}_{b_1} \right| \\ - 2 \cdot \left| \underbrace{(a_1)^2 (b_2)^2 (b_1)^1 (a_2)^1 (a_1^*)^0 (b_2^*)^1 (b_1^*)^1 (a_2^*)^0}_{b_1} \right| + 2 \cdot \left| \underbrace{(a_1)^2 (b_2)^2 (b_1)^1 (a_2)^1 (a_1^*)^1 (b_2^*)^0 (b_1^*)^0 (a_2^*)^1}_{b_1} \right| \\ + 2 \cdot \left| \underbrace{(a_1)^2 (b_2)^1 (b_1)^1 (a_2)^2 (a_1^*)^0 (b_2^*)^0 (b_1^*)^1 (a_2^*)^1}_{b_1} \right| - 2 \cdot \left| \underbrace{(a_1)^1 (b_2)^2 (b_1)^2 (a_2)^1 (a_1^*)^0 (b_2^*)^0 (b_1^*)^1 (a_2^*)^1}_{b_1} \right|$$

3.12 $e^3 a_1 * i^3 b_1 - i^3 b_1 * e^3 a_1$

$$e^3 a_1 * i^3 b_1 - i^3 b_1 * e^3 a_1$$

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} \\ + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix}$$

monomer ci vectors:

$$+ \left| \underbrace{(b_2^l)^1 (b_2^{*l})^1 (a_2^l)^2 (a_2^{*l})^0}_{a_1^l} \right| \cdot \left| \underbrace{(b_2^r)^2 (b_2^{*r})^1 (a_2^r)^1 (a_2^{*r})^0}_{b_1^r} \right| - \left| \underbrace{(b_2^l)^2 (b_2^{*l})^1 (a_2^l)^1 (a_2^{*l})^0}_{b_1^l} \right| \cdot \left| \underbrace{(b_2^r)^1 (b_2^{*r})^1 (a_2^r)^2 (a_2^{*r})^0}_{a_1^r} \right| \\ - \left| \underbrace{(b_2^l)^1 (b_2^{*l})^1 (a_2^l)^2 (a_2^{*l})^0}_{a_1^l} \right| \cdot \left| \underbrace{(b_2^r)^1 (b_2^{*r})^0 (a_2^r)^2 (a_2^{*r})^1}_{b_1^r} \right| + \left| \underbrace{(b_2^l)^1 (b_2^{*l})^0 (a_2^l)^2 (a_2^{*l})^1}_{b_1^l} \right| \cdot \left| \underbrace{(b_2^r)^1 (b_2^{*r})^1 (a_2^r)^2 (a_2^{*r})^0}_{a_1^r} \right| \\ + \left| \underbrace{(b_2^l)^2 (b_2^{*l})^0 (a_2^l)^1 (a_2^{*l})^1}_{a_1^l} \right| \cdot \left| \underbrace{(b_2^r)^2 (b_2^{*r})^1 (a_2^r)^1 (a_2^{*r})^0}_{b_1^r} \right| - \left| \underbrace{(b_2^l)^2 (b_2^{*l})^1 (a_2^l)^1 (a_2^{*l})^0}_{b_1^l} \right| \cdot \left| \underbrace{(b_2^r)^2 (b_2^{*r})^0 (a_2^r)^1 (a_2^{*r})^1}_{a_1^r} \right| \\ - \left| \underbrace{(b_2^l)^2 (b_2^{*l})^0 (a_2^l)^1 (a_2^{*l})^1}_{a_1^l} \right| \cdot \left| \underbrace{(b_2^r)^1 (b_2^{*r})^0 (a_2^r)^2 (a_2^{*r})^1}_{b_1^r} \right| + \left| \underbrace{(b_2^l)^1 (b_2^{*l})^0 (a_2^l)^2 (a_2^{*l})^1}_{b_1^l} \right| \cdot \left| \underbrace{(b_2^r)^2 (b_2^{*r})^0 (a_2^r)^1 (a_2^{*r})^1}_{a_1^r} \right|$$

substitution by dimer orbitals:

$$+ \left| \underbrace{(+a_1 + b_2)^1 (+a_1^* + b_2^*)^1 (+b_1 + a_2)^2 (+b_1^* + a_2^*)^0}_{a_1^l} \right| \cdot \left| \underbrace{(-a_1 + b_2)^2 (-a_1^* + b_2^*)^1 (-b_1 + a_2)^1 (-b_1^* + a_2^*)^0}_{b_1^r} \right| + \dots$$

(multiplied out and summed up) dimer ci vectors:

$$-2 \cdot \left| \underbrace{(a_1)^1 (b_2)^2 (b_1)^1 (a_2)^2 (a_1^*)^1 (b_2^*)^1 (b_1^*)^0 (a_2^*)^0}_{a_2} \right| + 2 \cdot \left| \underbrace{(a_1)^2 (b_2)^1 (b_1)^2 (a_2)^1 (a_1^*)^1 (b_2^*)^1 (b_1^*)^0 (a_2^*)^0}_{a_2} \right| \\ + 2 \cdot \left| \underbrace{(a_1)^1 (b_2)^1 (b_1)^2 (a_2)^2 (a_1^*)^1 (b_2^*)^0 (b_1^*)^1 (a_2^*)^0}_{a_2} \right| - 2 \cdot \left| \underbrace{(a_1)^1 (b_2)^1 (b_1)^2 (a_2)^2 (a_1^*)^0 (b_2^*)^1 (b_1^*)^0 (a_2^*)^1}_{a_2} \right| \\ - 2 \cdot \left| \underbrace{(a_1)^2 (b_2)^2 (b_1)^1 (a_2)^1 (a_1^*)^0 (b_2^*)^1 (b_1^*)^0 (a_2^*)^1}_{a_2} \right| + 2 \cdot \left| \underbrace{(a_1)^2 (b_2)^2 (b_1)^1 (a_2)^1 (a_1^*)^1 (b_2^*)^0 (b_1^*)^1 (a_2^*)^0}_{a_2} \right| \\ + 2 \cdot \left| \underbrace{(a_1)^2 (b_2)^1 (b_1)^2 (a_2)^1 (a_1^*)^0 (b_2^*)^0 (b_1^*)^1 (a_2^*)^1}_{a_2} \right| - 2 \cdot \left| \underbrace{(a_1)^1 (b_2)^2 (b_1)^1 (a_2)^2 (a_1^*)^0 (b_2^*)^0 (b_1^*)^1 (a_2^*)^1}_{a_2} \right|$$

3.13 $e^3 a_1 * e^3 b_1 + e^3 b_1 * e^3 a_1$

$$e^3 a_1 * e^3 b_1 + e^3 b_1 * e^3 a_1$$

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} \\ + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix}$$

monomer ci vectors:

$$+ \left| \underbrace{(b_2^l)^1 (b_2^{*l})^1 (a_2^l)^2 (a_2^{*l})^0}_{a_1^l} \right| \cdot \left| \underbrace{(b_2^r)^2 (b_2^{*r})^1 (a_2^r)^1 (a_2^{*r})^0}_{b_1^r} \right| + \left| \underbrace{(b_2^l)^2 (b_2^{*l})^1 (a_2^l)^1 (a_2^{*l})^0}_{b_1^l} \right| \cdot \left| \underbrace{(b_2^r)^1 (b_2^{*r})^1 (a_2^r)^2 (a_2^{*r})^0}_{a_1^r} \right| \\ + \left| \underbrace{(b_2^l)^1 (b_2^{*l})^1 (a_2^l)^2 (a_2^{*l})^0}_{a_1^l} \right| \cdot \left| \underbrace{(b_2^r)^1 (b_2^{*r})^0 (a_2^r)^2 (a_2^{*r})^1}_{b_1^r} \right| + \left| \underbrace{(b_2^l)^1 (b_2^{*l})^0 (a_2^l)^2 (a_2^{*l})^1}_{b_1^l} \right| \cdot \left| \underbrace{(b_2^r)^1 (b_2^{*r})^1 (a_2^r)^2 (a_2^{*r})^0}_{a_1^r} \right| \\ + \left| \underbrace{(b_2^l)^2 (b_2^{*l})^0 (a_2^l)^1 (a_2^{*l})^1}_{a_1^l} \right| \cdot \left| \underbrace{(b_2^r)^2 (b_2^{*r})^1 (a_2^r)^1 (a_2^{*r})^0}_{b_1^r} \right| + \left| \underbrace{(b_2^l)^2 (b_2^{*l})^1 (a_2^l)^1 (a_2^{*l})^0}_{b_1^l} \right| \cdot \left| \underbrace{(b_2^r)^2 (b_2^{*r})^0 (a_2^r)^1 (a_2^{*r})^1}_{a_1^r} \right| \\ + \left| \underbrace{(b_2^l)^2 (b_2^{*l})^0 (a_2^l)^1 (a_2^{*l})^1}_{a_1^l} \right| \cdot \left| \underbrace{(b_2^r)^1 (b_2^{*r})^0 (a_2^r)^2 (a_2^{*r})^1}_{b_1^r} \right| + \left| \underbrace{(b_2^l)^1 (b_2^{*l})^0 (a_2^l)^2 (a_2^{*l})^1}_{b_1^l} \right| \cdot \left| \underbrace{(b_2^r)^2 (b_2^{*r})^0 (a_2^r)^1 (a_2^{*r})^1}_{a_1^r} \right|$$

substitution by dimer orbitals:

$$+ \left| \underbrace{(+a_1 + b_2)^1 (+a_1^* + b_2^*)^1 (+b_1 + a_2)^2 (+b_1^* + a_2^*)^0}_{a_1^l} \right| \cdot \left| \underbrace{(-a_1 + b_2)^2 (-a_1^* + b_2^*)^1 (-b_1 + a_2)^1 (-b_1^* + a_2^*)^0}_{b_1^r} \right| + \dots$$

(multiplied out and summed up) dimer ci vectors:

$$+2 \cdot \left| \underbrace{(a_1)^1 (b_2)^2 (b_1)^2 (a_2)^1 (a_1^*)^1 (b_2^*)^1 (b_1^*)^0 (a_2^*)^0}_{b_1} \right| - 2 \cdot \left| \underbrace{(a_1)^2 (b_2)^1 (b_1)^1 (a_2)^2 (a_1^*)^1 (b_2^*)^1 (b_1^*)^0 (a_2^*)^0}_{b_1} \right| \\ +2 \cdot \left| \underbrace{(a_1)^1 (b_2)^1 (b_1)^2 (a_2)^2 (a_1^*)^1 (b_2^*)^0 (b_1^*)^0 (a_2^*)^1}_{b_1} \right| - 2 \cdot \left| \underbrace{(a_1)^1 (b_2)^1 (b_1)^2 (a_2)^2 (a_1^*)^0 (b_2^*)^1 (b_1^*)^1 (a_2^*)^0}_{b_1} \right| \\ -2 \cdot \left| \underbrace{(a_1)^2 (b_2)^2 (b_1)^1 (a_2)^1 (a_1^*)^0 (b_2^*)^1 (b_1^*)^1 (a_2^*)^0}_{b_1} \right| + 2 \cdot \left| \underbrace{(a_1)^2 (b_2)^2 (b_1)^1 (a_2)^1 (a_1^*)^1 (b_2^*)^0 (b_1^*)^0 (a_2^*)^1}_{b_1} \right| \\ -2 \cdot \left| \underbrace{(a_1)^2 (b_2)^1 (b_1)^1 (a_2)^2 (a_1^*)^0 (b_2^*)^0 (b_1^*)^1 (a_2^*)^1}_{b_1} \right| + 2 \cdot \left| \underbrace{(a_1)^1 (b_2)^2 (b_1)^2 (a_2)^1 (a_1^*)^0 (b_2^*)^0 (b_1^*)^1 (a_2^*)^1}_{b_1} \right|$$

3.14 $e^3 a_1 * e^3 b_1 - e^3 b_1 * e^3 a_1$

$$e^3 a_1 * e^3 b_1 - e^3 b_1 * e^3 a_1$$

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} \\ + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix}$$

monomer ci vectors:

$$+ \left| \underbrace{(b_2^l)^1 (b_2^{*l})^1 (a_2^l)^2 (a_2^{*l})^0}_{a_1^l} \cdot \underbrace{(b_2^r)^2 (b_2^{*r})^1 (a_2^r)^1 (a_2^{*r})^0}_{b_1^r} \right| - \left| \underbrace{(b_2^l)^2 (b_2^{*l})^1 (a_2^l)^1 (a_2^{*l})^0}_{b_1^l} \cdot \underbrace{(b_2^r)^1 (b_2^{*r})^1 (a_2^r)^2 (a_2^{*r})^0}_{a_1^r} \right| \\ + \left| \underbrace{(b_2^l)^1 (b_2^{*l})^1 (a_2^l)^2 (a_2^{*l})^0}_{a_1^l} \cdot \underbrace{(b_2^r)^1 (b_2^{*r})^0 (a_2^r)^2 (a_2^{*r})^1}_{b_1^r} \right| - \left| \underbrace{(b_2^l)^1 (b_2^{*l})^0 (a_2^l)^2 (a_2^{*l})^1}_{b_1^l} \cdot \underbrace{(b_2^r)^1 (b_2^{*r})^1 (a_2^r)^2 (a_2^{*r})^0}_{a_1^r} \right| \\ + \left| \underbrace{(b_2^l)^2 (b_2^{*l})^0 (a_2^l)^1 (a_2^{*l})^1}_{a_1^l} \cdot \underbrace{(b_2^r)^2 (b_2^{*r})^1 (a_2^r)^1 (a_2^{*r})^0}_{b_1^r} \right| - \left| \underbrace{(b_2^l)^2 (b_2^{*l})^1 (a_2^l)^1 (a_2^{*l})^0}_{b_1^l} \cdot \underbrace{(b_2^r)^2 (b_2^{*r})^0 (a_2^r)^1 (a_2^{*r})^1}_{a_1^r} \right| \\ + \left| \underbrace{(b_2^l)^2 (b_2^{*l})^0 (a_2^l)^1 (a_2^{*l})^1}_{a_1^l} \cdot \underbrace{(b_2^r)^1 (b_2^{*r})^0 (a_2^r)^2 (a_2^{*r})^1}_{b_1^r} \right| - \left| \underbrace{(b_2^l)^1 (b_2^{*l})^0 (a_2^l)^2 (a_2^{*l})^1}_{b_1^l} \cdot \underbrace{(b_2^r)^2 (b_2^{*r})^0 (a_2^r)^1 (a_2^{*r})^1}_{a_1^r} \right|$$

substitution by dimer orbitals:

$$+ \left| \underbrace{(+a_1 + b_2)^1 (+a_1^* + b_2^*)^1 (+b_1 + a_2)^2 (+b_1^* + a_2^*)^0}_{a_1^l} \right| \cdot \left| \underbrace{(-a_1 + b_2)^2 (-a_1^* + b_2^*)^1 (-b_1 + a_2)^1 (-b_1^* + a_2^*)^0}_{b_1^r} \right| + \dots$$

(multiplied out and summed up) dimer ci vectors:

$$-2 \cdot \left| \underbrace{(a_1)^1 (b_2)^2 (b_1)^1 (a_2)^2 (a_1^*)^1 (b_2^*)^1 (b_1^*)^0 (a_2^*)^0}_{a_2} \right| + 2 \cdot \left| \underbrace{(a_1)^2 (b_2)^1 (b_1)^2 (a_2)^1 (a_1^*)^1 (b_2^*)^1 (b_1^*)^0 (a_2^*)^0}_{a_2} \right| \\ -2 \cdot \left| \underbrace{(a_1)^1 (b_2)^1 (b_1)^2 (a_2)^2 (a_1^*)^1 (b_2^*)^0 (b_1^*)^1 (a_2^*)^0}_{a_2} \right| + 2 \cdot \left| \underbrace{(a_1)^1 (b_2)^1 (b_1)^2 (a_2)^2 (a_1^*)^0 (b_2^*)^1 (b_1^*)^0 (a_2^*)^1}_{a_2} \right| \\ -2 \cdot \left| \underbrace{(a_1)^2 (b_2)^2 (b_1)^1 (a_2)^1 (a_1^*)^0 (b_2^*)^1 (b_1^*)^0 (a_2^*)^1}_{a_2} \right| + 2 \cdot \left| \underbrace{(a_1)^2 (b_2)^2 (b_1)^1 (a_2)^1 (a_1^*)^1 (b_2^*)^0 (b_1^*)^1 (a_2^*)^0}_{a_2} \right| \\ -2 \cdot \left| \underbrace{(a_1)^2 (b_2)^1 (b_1)^2 (a_2)^1 (a_1^*)^0 (b_2^*)^0 (b_1^*)^1 (a_2^*)^1}_{a_2} \right| + 2 \cdot \left| \underbrace{(a_1)^1 (b_2)^2 (b_1)^1 (a_2)^2 (a_1^*)^0 (b_2^*)^0 (b_1^*)^1 (a_2^*)^1}_{a_2} \right|$$

3.15 $i^3 b_1 * i^3 b_1$

$$i^3 b_1 * i^3 b_1$$

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \uparrow\uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \uparrow\uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow\uparrow & \uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow\uparrow & \uparrow & \uparrow\uparrow & \uparrow \end{pmatrix}$$

monomer ci vectors:

$$+ \left| \underbrace{(b_2^l)^2 (b_2^{*l})^1 (a_2^l)^1 (a_2^{*l})^0}_{b_1^l} \right| \cdot \left| \underbrace{(b_2^r)^2 (b_2^{*r})^1 (a_2^r)^1 (a_2^{*r})^0}_{b_1^r} \right| - \left| \underbrace{(b_2^l)^2 (b_2^{*l})^1 (a_2^l)^1 (a_2^{*l})^0}_{b_1^l} \right| \cdot \left| \underbrace{(b_2^r)^1 (b_2^{*r})^0 (a_2^r)^2 (a_2^{*r})^1}_{b_1^r} \right| \\ - \left| \underbrace{(b_2^l)^1 (b_2^{*l})^0 (a_2^l)^2 (a_2^{*l})^1}_{b_1^l} \right| \cdot \left| \underbrace{(b_2^r)^2 (b_2^{*r})^1 (a_2^r)^1 (a_2^{*r})^0}_{b_1^r} \right| + \left| \underbrace{(b_2^l)^1 (b_2^{*l})^0 (a_2^l)^2 (a_2^{*l})^1}_{b_1^l} \right| \cdot \left| \underbrace{(b_2^r)^1 (b_2^{*r})^0 (a_2^r)^2 (a_2^{*r})^1}_{b_1^r} \right|$$

substitution by dimer orbitals:

$$+ \left| \underbrace{(+a_1 + b_2)^2 (+a_1^* + b_2^*)^1 (+b_1 + a_2)^1 (+b_1^* + a_2^*)^0}_{b_1^l} \right| \cdot \left| \underbrace{(-a_1 + b_2)^2 (-a_1^* + b_2^*)^1 (-b_1 + a_2)^1 (-b_1^* + a_2^*)^0}_{b_1^r} \right| + \dots$$

(multiplied out and summed up) dimer ci vectors:

$$+ \left| \underbrace{(a_1)^2 (b_2)^2 (b_1)^1 (a_2)^1 (a_1^*)^1 (b_2^*)^1 (b_1^*)^0 (a_2^*)^0}_{a_1} \right| + 2 \cdot \left| \underbrace{(a_1)^2 (b_2)^1 (b_1)^1 (a_2)^2 (a_1^*)^1 (b_2^*)^0 (b_1^*)^0 (a_2^*)^1}_{a_1} \right| \\ - 2 \cdot \left| \underbrace{(a_1)^2 (b_2)^1 (b_1)^2 (a_2)^1 (a_1^*)^1 (b_2^*)^0 (b_1^*)^1 (a_2^*)^0}_{a_1} \right| - 2 \cdot \left| \underbrace{(a_1)^2 (b_2)^1 (b_1)^1 (a_2)^2 (a_1^*)^0 (b_2^*)^1 (b_1^*)^1 (a_2^*)^0}_{a_1} \right| \\ + 2 \cdot \left| \underbrace{(a_1)^2 (b_2)^1 (b_1)^2 (a_2)^1 (a_1^*)^0 (b_2^*)^1 (b_1^*)^0 (a_2^*)^1}_{a_1} \right| + 2 \cdot \left| \underbrace{(a_1)^1 (b_2)^2 (b_1)^1 (a_2)^2 (a_1^*)^1 (b_2^*)^0 (b_1^*)^1 (a_2^*)^0}_{a_1} \right| \\ - 2 \cdot \left| \underbrace{(a_1)^1 (b_2)^2 (b_1)^2 (a_2)^1 (a_1^*)^1 (b_2^*)^0 (b_1^*)^0 (a_2^*)^1}_{a_1} \right| - 2 \cdot \left| \underbrace{(a_1)^1 (b_2)^2 (b_1)^1 (a_2)^2 (a_1^*)^0 (b_2^*)^1 (b_1^*)^0 (a_2^*)^1}_{a_1} \right| \\ + 2 \cdot \left| \underbrace{(a_1)^1 (b_2)^2 (b_1)^2 (a_2)^1 (a_1^*)^0 (b_2^*)^1 (b_1^*)^1 (a_2^*)^0}_{a_1} \right| + \left| \underbrace{(a_1)^1 (b_2)^1 (b_1)^2 (a_2)^2 (a_1^*)^0 (b_2^*)^0 (b_1^*)^1 (a_2^*)^1}_{a_1} \right|$$

3.16 $i^3 b_1 * e^3 b_1 + e^3 b_1 * i^3 b_1$

$$i^3 b_1 * e^3 b_1 + e^3 b_1 * i^3 b_1$$

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \uparrow\uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \uparrow\uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \uparrow\uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow\uparrow & \uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} \\ - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow\uparrow & \uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \uparrow\uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow\uparrow & \uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow\uparrow & \uparrow & \uparrow\uparrow & \uparrow \end{pmatrix}$$

monomer ci vectors:

$$+ \left| \underbrace{(b_2^l)^2 (b_2^{*l})^1 (a_2^l)^1 (a_2^{*l})^0}_{b_1^l} \right| \cdot \left| \underbrace{(b_2^r)^2 (b_2^{*r})^1 (a_2^r)^1 (a_2^{*r})^0}_{b_1^r} \right| + \left| \underbrace{(b_2^l)^2 (b_2^{*l})^1 (a_2^l)^1 (a_2^{*l})^0}_{b_1^l} \right| \cdot \left| \underbrace{(b_2^r)^2 (b_2^{*r})^1 (a_2^r)^1 (a_2^{*r})^0}_{b_1^r} \right| \\ + \left| \underbrace{(b_2^l)^2 (b_2^{*l})^1 (a_2^l)^1 (a_2^{*l})^0}_{b_1^l} \right| \cdot \left| \underbrace{(b_2^r)^1 (b_2^{*r})^0 (a_2^r)^2 (a_2^{*r})^1}_{b_1^r} \right| + \left| \underbrace{(b_2^l)^1 (b_2^{*l})^0 (a_2^l)^2 (a_2^{*l})^1}_{b_1^l} \right| \cdot \left| \underbrace{(b_2^r)^2 (b_2^{*r})^1 (a_2^r)^1 (a_2^{*r})^0}_{b_1^r} \right| \\ - \left| \underbrace{(b_2^l)^1 (b_2^{*l})^0 (a_2^l)^2 (a_2^{*l})^1}_{b_1^l} \right| \cdot \left| \underbrace{(b_2^r)^2 (b_2^{*r})^1 (a_2^r)^1 (a_2^{*r})^0}_{b_1^r} \right| - \left| \underbrace{(b_2^l)^2 (b_2^{*l})^1 (a_2^l)^1 (a_2^{*l})^0}_{b_1^l} \right| \cdot \left| \underbrace{(b_2^r)^1 (b_2^{*r})^0 (a_2^r)^2 (a_2^{*r})^1}_{b_1^r} \right| \\ - \left| \underbrace{(b_2^l)^1 (b_2^{*l})^0 (a_2^l)^2 (a_2^{*l})^1}_{b_1^l} \right| \cdot \left| \underbrace{(b_2^r)^1 (b_2^{*r})^0 (a_2^r)^2 (a_2^{*r})^1}_{b_1^r} \right|$$

substitution by dimer orbitals:

$$+ \left| \underbrace{(+a_1 + b_2)^2 (+a_1^* + b_2^*)^1 (+b_1 + a_2)^1 (+b_1^* + a_2^*)^0}_{b_1^l} \right| \cdot \left| \underbrace{(-a_1 + b_2)^2 (-a_1^* + b_2^*)^1 (-b_1 + a_2)^1 (-b_1^* + a_2^*)^0}_{b_1^r} \right| + \dots$$

(multiplied out and summed up) dimer ci vectors:

$$+ 2 \cdot \left| \underbrace{(a_1)^2 (b_2)^2 (b_1)^1 (a_2)^1 (a_1^*)^1 (b_2^*)^1 (b_1^*)^0 (a_2^*)^0}_{a_1} \right| - 2 \cdot \left| \underbrace{(a_1)^1 (b_2)^1 (b_1)^2 (a_2)^2 (a_1^*)^0 (b_2^*)^0 (b_1^*)^1 (a_2^*)^1}_{a_1} \right|$$

3.17 $i^3 b_1 * e^3 b_1 - e^3 b_1 * i^3 b_1$

$$i^3 b_1 * e^3 b_1 - e^3 b_1 * i^3 b_1$$

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \uparrow\uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \uparrow\uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \uparrow\uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow\uparrow & \uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} \\ - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow\uparrow & \uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \uparrow\uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow\uparrow & \uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow\uparrow & \uparrow & \uparrow\uparrow & \uparrow \end{pmatrix}$$

monomer ci vectors:

$$+ \left| \underbrace{(b_2^l)^2 (b_2^{*l})^1 (a_2^l)^1 (a_2^{*l})^0}_{b_1^l} \right| \cdot \left| \underbrace{(b_2^r)^2 (b_2^{*r})^1 (a_2^r)^1 (a_2^{*r})^0}_{b_1^r} \right| - \left| \underbrace{(b_2^l)^2 (b_2^{*l})^1 (a_2^l)^1 (a_2^{*l})^0}_{b_1^l} \right| \cdot \left| \underbrace{(b_2^r)^2 (b_2^{*r})^1 (a_2^r)^1 (a_2^{*r})^0}_{b_1^r} \right| \\ + \left| \underbrace{(b_2^l)^2 (b_2^{*l})^1 (a_2^l)^1 (a_2^{*l})^0}_{b_1^l} \right| \cdot \left| \underbrace{(b_2^r)^1 (b_2^{*r})^0 (a_2^r)^2 (a_2^{*r})^1}_{b_1^r} \right| - \left| \underbrace{(b_2^l)^1 (b_2^{*l})^0 (a_2^l)^2 (a_2^{*l})^1}_{b_1^l} \right| \cdot \left| \underbrace{(b_2^r)^2 (b_2^{*r})^1 (a_2^r)^1 (a_2^{*r})^0}_{b_1^r} \right| \\ - \left| \underbrace{(b_2^l)^1 (b_2^{*l})^0 (a_2^l)^2 (a_2^{*l})^1}_{b_1^l} \right| \cdot \left| \underbrace{(b_2^r)^2 (b_2^{*r})^1 (a_2^r)^1 (a_2^{*r})^0}_{b_1^r} \right| + \left| \underbrace{(b_2^l)^2 (b_2^{*l})^1 (a_2^l)^1 (a_2^{*l})^0}_{b_1^l} \right| \cdot \left| \underbrace{(b_2^r)^1 (b_2^{*r})^0 (a_2^r)^2 (a_2^{*r})^1}_{b_1^r} \right| \\ - \left| \underbrace{(b_2^l)^1 (b_2^{*l})^0 (a_2^l)^2 (a_2^{*l})^1}_{b_1^l} \right| \cdot \left| \underbrace{(b_2^r)^1 (b_2^{*r})^0 (a_2^r)^2 (a_2^{*r})^1}_{b_1^r} \right| + \left| \underbrace{(b_2^l)^1 (b_2^{*l})^0 (a_2^l)^2 (a_2^{*l})^1}_{b_1^l} \right| \cdot \left| \underbrace{(b_2^r)^1 (b_2^{*r})^0 (a_2^r)^2 (a_2^{*r})^1}_{b_1^r} \right|$$

substitution by dimer orbitals:

$$+ \left| \underbrace{(+a_1 + b_2)^2 (+a_1^* + b_2^*)^1 (+b_1 + a_2)^1 (+b_1^* + a_2^*)^0}_{b_1^l} \right| \cdot \left| \underbrace{(-a_1 + b_2)^2 (-a_1^* + b_2^*)^1 (-b_1 + a_2)^1 (-b_1^* + a_2^*)^0}_{b_1^r} \right| + \dots$$

(multiplied out and summed up) dimer ci vectors:

$$+4 \cdot \left| \underbrace{(a_1)^2 (b_2)^1 (b_1)^1 (a_2)^2 (a_1^*)^1 (b_2^*)^0 (b_1^*)^1 (a_2^*)^0}_{b_2} \right| - 4 \cdot \left| \underbrace{(a_1)^2 (b_2)^1 (b_1)^2 (a_2)^1 (a_1^*)^1 (b_2^*)^0 (b_1^*)^0 (a_2^*)^1}_{b_2} \right| \\ - 4 \cdot \left| \underbrace{(a_1)^2 (b_2)^1 (b_1)^1 (a_2)^2 (a_1^*)^0 (b_2^*)^1 (b_1^*)^0 (a_2^*)^1}_{b_2} \right| + 4 \cdot \left| \underbrace{(a_1)^2 (b_2)^1 (b_1)^2 (a_2)^1 (a_1^*)^0 (b_2^*)^1 (b_1^*)^1 (a_2^*)^0}_{b_2} \right| \\ + 4 \cdot \left| \underbrace{(a_1)^1 (b_2)^2 (b_1)^1 (a_2)^2 (a_1^*)^1 (b_2^*)^0 (b_1^*)^0 (a_2^*)^1}_{b_2} \right| - 4 \cdot \left| \underbrace{(a_1)^1 (b_2)^2 (b_1)^2 (a_2)^1 (a_1^*)^1 (b_2^*)^0 (b_1^*)^1 (a_2^*)^0}_{b_2} \right| \\ - 4 \cdot \left| \underbrace{(a_1)^1 (b_2)^2 (b_1)^1 (a_2)^2 (a_1^*)^0 (b_2^*)^1 (b_1^*)^1 (a_2^*)^0}_{b_2} \right| + 4 \cdot \left| \underbrace{(a_1)^1 (b_2)^2 (b_1)^2 (a_2)^1 (a_1^*)^0 (b_2^*)^1 (b_1^*)^0 (a_2^*)^1}_{b_2} \right|$$

3.18 $e^3 b_1 * e^3 b_1$

$$e^3 b_1 * e^3 b_1$$

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \uparrow\uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \uparrow\uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow\uparrow & \uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow\uparrow & \uparrow & \uparrow\uparrow & \uparrow \end{pmatrix}$$

monomer ci vectors:

$$+ \left| \underbrace{(b_2^l)^2 (b_2^{*l})^1 (a_2^l)^1 (a_2^{*l})^0}_{b_1^l} \right| \cdot \left| \underbrace{(b_2^r)^2 (b_2^{*r})^1 (a_2^r)^1 (a_2^{*r})^0}_{b_1^r} \right| + \left| \underbrace{(b_2^l)^2 (b_2^{*l})^1 (a_2^l)^1 (a_2^{*l})^0}_{b_1^l} \right| \cdot \left| \underbrace{(b_2^r)^1 (b_2^{*r})^0 (a_2^r)^2 (a_2^{*r})^1}_{b_1^r} \right|$$

$$+ \left| \underbrace{(b_2^l)^1 (b_2^{*l})^0 (a_2^l)^2 (a_2^{*l})^1}_{b_1^l} \right| \cdot \left| \underbrace{(b_2^r)^2 (b_2^{*r})^1 (a_2^r)^1 (a_2^{*r})^0}_{b_1^r} \right| + \left| \underbrace{(b_2^l)^1 (b_2^{*l})^0 (a_2^l)^2 (a_2^{*l})^1}_{b_1^l} \right| \cdot \left| \underbrace{(b_2^r)^1 (b_2^{*r})^0 (a_2^r)^2 (a_2^{*r})^1}_{b_1^r} \right|$$

substitution by dimer orbitals:

$$+ \left| \underbrace{(+a_1 + b_2)^2 (+a_1^* + b_2^*)^1 (+b_1 + a_2)^1 (+b_1^* + a_2^*)^0}_{b_1^l} \right| \cdot \left| \underbrace{(-a_1 + b_2)^2 (-a_1^* + b_2^*)^1 (-b_1 + a_2)^1 (-b_1^* + a_2^*)^0}_{b_1^r} \right| + \dots$$

(multiplied out and summed up) dimer ci vectors:

$$+ \left| \underbrace{(a_1)^2 (b_2)^2 (b_1)^1 (a_2)^1 (a_1^*)^1 (b_2^*)^1 (b_1^*)^0 (a_2^*)^0}_{a_1} \right| - 2 \cdot \left| \underbrace{(a_1)^2 (b_2)^1 (b_1)^1 (a_2)^2 (a_1^*)^1 (b_2^*)^0 (b_1^*)^0 (a_2^*)^1}_{a_1} \right|$$

$$+ 2 \cdot \left| \underbrace{(a_1)^2 (b_2)^1 (b_1)^2 (a_2)^1 (a_1^*)^1 (b_2^*)^0 (b_1^*)^1 (a_2^*)^0}_{a_1} \right| + 2 \cdot \left| \underbrace{(a_1)^2 (b_2)^1 (b_1)^1 (a_2)^2 (a_1^*)^0 (b_2^*)^1 (b_1^*)^1 (a_2^*)^0}_{a_1} \right|$$

$$- 2 \cdot \left| \underbrace{(a_1)^2 (b_2)^1 (b_1)^2 (a_2)^1 (a_1^*)^0 (b_2^*)^1 (b_1^*)^0 (a_2^*)^1}_{a_1} \right| - 2 \cdot \left| \underbrace{(a_1)^1 (b_2)^2 (b_1)^1 (a_2)^2 (a_1^*)^1 (b_2^*)^0 (b_1^*)^1 (a_2^*)^0}_{a_1} \right|$$

$$+ 2 \cdot \left| \underbrace{(a_1)^1 (b_2)^2 (b_1)^2 (a_2)^1 (a_1^*)^1 (b_2^*)^0 (b_1^*)^0 (a_2^*)^1}_{a_1} \right| + 2 \cdot \left| \underbrace{(a_1)^1 (b_2)^2 (b_1)^1 (a_2)^2 (a_1^*)^0 (b_2^*)^1 (b_1^*)^0 (a_2^*)^1}_{a_1} \right|$$

$$- 2 \cdot \left| \underbrace{(a_1)^1 (b_2)^2 (b_1)^2 (a_2)^1 (a_1^*)^0 (b_2^*)^1 (b_1^*)^1 (a_2^*)^0}_{a_1} \right| + \left| \underbrace{(a_1)^1 (b_2)^1 (b_1)^2 (a_2)^2 (a_1^*)^0 (b_2^*)^0 (b_1^*)^1 (a_2^*)^1}_{a_1} \right|$$

4 Symmetry Conclusion

Dimer State	Symmetry
$S * Q + Q * S$	a_1
$S * Q - Q * S$	b_2
$i^3 a_1 * i^3 a_1$	a_1
$i^3 a_1 * e^3 a_1 + e^3 a_1 * i^3 a_1$	a_1
$i^3 a_1 * e^3 a_1 - e^3 a_1 * i^3 a_1$	b_2
$i^3 a_1 * i^3 b_1 + i^3 b_1 * i^3 a_1$	b_1
$i^3 a_1 * i^3 b_1 - i^3 b_1 * i^3 a_1$	a_2
$i^3 a_1 * e^3 b_1 + e^3 b_1 * i^3 a_1$	b_1
$i^3 a_1 * e^3 b_1 - e^3 b_1 * i^3 a_1$	a_2
$e^3 a_1 * e^3 a_1$	a_1
$e^3 a_1 * i^3 b_1 + i^3 b_1 * e^3 a_1$	b_1
$e^3 a_1 * i^3 b_1 - i^3 b_1 * e^3 a_1$	a_2
$e^3 a_1 * e^3 b_1 + e^3 b_1 * e^3 a_1$	b_1
$e^3 a_1 * e^3 b_1 - e^3 b_1 * e^3 a_1$	a_2
$i^3 b_1 * i^3 b_1$	a_1
$i^3 b_1 * e^3 b_1 + e^3 b_1 * i^3 b_1$	a_1
$i^3 b_1 * e^3 b_1 - e^3 b_1 * i^3 b_1$	b_2
$e^3 b_1 * e^3 b_1$	a_1

$\Rightarrow$ 7x a_1 and 3x b_2 and 4x b_1 and 4x a_2

5 CI Vector Conclusion

Dimer States and Their CI Vectors:

- $S * Q + Q * S$:

$$\begin{aligned} & -2 \cdot |2a2a0a0a| - 2 \cdot |2aa20aa0| - 2 \cdot |2a2aa0a0| - 2 \cdot |2aa2a00a| - 2 \cdot |a22a0aa0| - 2 \cdot |a2a20a0a| \\ & -2 \cdot |a22aa00a| - 2 \cdot |a2a2a0a0| \end{aligned}$$

- $S * Q - Q * S$:

$$\begin{aligned} & +2 \cdot |2a2a0a0a| + 2 \cdot |2aa20aa0| + 2 \cdot |2a2aa0a0| + 2 \cdot |2aa2a00a| + 2 \cdot |a22a0aa0| + 2 \cdot |a2a20a0a| \\ & +2 \cdot |a22aa00a| + 2 \cdot |a2a2a0a0| \end{aligned}$$

- $i^3 a_1 * i^3 a_1$:

$$\begin{aligned} & + |aa22aa00| + 2 \cdot |a22aa00a| + 2 \cdot |a2a2a0a0| - 2 \cdot |a22a0aa0| - 2 \cdot |a2a20a0a| - 2 \cdot |2a2aa0a0| \\ & -2 \cdot |2aa2a00a| + 2 \cdot |2a2a0a0a| + 2 \cdot |2aa20aa0| + |22aa00aa| \end{aligned}$$

- $i^3 a_1 * e^3 a_1 + e^3 a_1 * i^3 a_1$:

$$+2 \cdot |aa22aa00| - 2 \cdot |22aa00aa|$$

- $i^3 a_1 * e^3 a_1 - e^3 a_1 * i^3 a_1$:

$$\begin{aligned} & +4 \cdot |a22aa0a0| + 4 \cdot |a2a2a00a| - 4 \cdot |a22a0aa0| - 4 \cdot |a2a20aa0| - 4 \cdot |2a2aa00a| - 4 \cdot |2aa2a0a0| \\ & +4 \cdot |2a2a0aa0| + 4 \cdot |2aa20a0a| \end{aligned}$$

- $i^3 a_1 * i^3 b_1 + i^3 b_1 * i^3 a_1$:

$$\begin{aligned} & +2 \cdot |a22aaa00| - 2 \cdot |2aa2aa00| - 2 \cdot |aa22a00a| + 2 \cdot |aa220aa0| + 2 \cdot |22aa0aa0| - 2 \cdot |22aaa00a| \\ & -2 \cdot |2aa200aa| + 2 \cdot |a22a00aa| \end{aligned}$$

- $i^3 a_1 * i^3 b_1 - i^3 b_1 * i^3 a_1$:

$$\begin{aligned} & -2 \cdot |a2a2aa00| + 2 \cdot |2a2aaa00| + 2 \cdot |aa22a0a0| - 2 \cdot |aa220a0a| + 2 \cdot |22aa0a0a| - 2 \cdot |22aaa0a0| \\ & -2 \cdot |2a2a00aa| + 2 \cdot |a2a200aa| \end{aligned}$$

- $i^3 a_1 * e^3 b_1 + e^3 b_1 * i^3 a_1$:

$$\begin{aligned} & +2 \cdot |a22aaa00| - 2 \cdot |2aa2aa00| + 2 \cdot |aa22a00a| - 2 \cdot |aa220aa0| + 2 \cdot |22aa0aa0| - 2 \cdot |22aaa00a| \\ & +2 \cdot |2aa200aa| - 2 \cdot |a22a00aa| \end{aligned}$$

- $i^3 a_1 * e^3 b_1 - e^3 b_1 * i^3 a_1$:

$$\begin{aligned} & -2 \cdot |a2a2aa00| + 2 \cdot |2a2aaa00| - 2 \cdot |aa22a0a0| + 2 \cdot |aa220a0a| + 2 \cdot |22aa0a0a| - 2 \cdot |22aaa0a0| \\ & +2 \cdot |2a2a00aa| - 2 \cdot |a2a200aa| \end{aligned}$$

- $e^3 a_1 * e^3 a_1$:

$$\begin{aligned} & + |aa22aa00| - 2 \cdot |a22aa00a| - 2 \cdot |a2a2a0a0| + 2 \cdot |a22a0aa0| + 2 \cdot |a2a20a0a| + 2 \cdot |2a2aa0a0| \\ & +2 \cdot |2aa2a00a| - 2 \cdot |2a2a0a0a| - 2 \cdot |2aa20aa0| + |22aa00aa| \end{aligned}$$

- $e^3 a_1 * i^3 b_1 + i^3 b_1 * e^3 a_1$:
 $+2 \cdot |a22aaa00| - 2 \cdot |2aa2aa00| - 2 \cdot |aa22a00a| + 2 \cdot |aa220aa0| - 2 \cdot |22aa0aa0| + 2 \cdot |22aaa00a|$
 $+2 \cdot |2aa200aa| - 2 \cdot |a22a00aa|$

- $e^3 a_1 * i^3 b_1 - i^3 b_1 * e^3 a_1$:
 $-2 \cdot |a2a2aa00| + 2 \cdot |2a2aaa00| + 2 \cdot |aa22a0a0| - 2 \cdot |aa220aa0| - 2 \cdot |22aa0aa0| + 2 \cdot |22aaa0a0|$
 $+2 \cdot |2a2a00aa| - 2 \cdot |a2a200aa|$

- $e^3 a_1 * e^3 b_1 + e^3 b_1 * e^3 a_1$:
 $+2 \cdot |a22aaa00| - 2 \cdot |2aa2aa00| + 2 \cdot |aa22a00a| - 2 \cdot |aa220aa0| - 2 \cdot |22aa0aa0| + 2 \cdot |22aaa00a|$
 $-2 \cdot |2aa200aa| + 2 \cdot |a22a00aa|$

- $e^3 a_1 * e^3 b_1 - e^3 b_1 * e^3 a_1$:
 $-2 \cdot |a2a2aa00| + 2 \cdot |2a2aaa00| - 2 \cdot |aa22a0a0| + 2 \cdot |aa220aa0| - 2 \cdot |22aa0aa0| + 2 \cdot |22aaa0a0|$
 $-2 \cdot |2a2a00aa| + 2 \cdot |a2a200aa|$

- $i^3 b_1 * i^3 b_1$:
 $+|22aaaa00| + 2 \cdot |2aa2a00a| - 2 \cdot |2a2aa0a0| - 2 \cdot |2aa20aa0| + 2 \cdot |2a2a0a0a| + 2 \cdot |a2a2a0a0|$
 $-2 \cdot |a22aa00a| - 2 \cdot |a2a20a0a| + 2 \cdot |a22a0aa0| + |aa2200aa|$

- $i^3 b_1 * e^3 b_1 + e^3 b_1 * i^3 b_1$:
 $+2 \cdot |22aaaa00| - 2 \cdot |aa2200aa|$

- $i^3 b_1 * e^3 b_1 - e^3 b_1 * i^3 b_1$:
 $+4 \cdot |2aa2a0a0| - 4 \cdot |2a2aa00a| - 4 \cdot |2aa20a0a| + 4 \cdot |2a2a0aa0| + 4 \cdot |a2a2a00a| - 4 \cdot |a22aa0a0|$
 $-4 \cdot |a2a20aa0| + 4 \cdot |a22a0a0a|$

- $e^3 b_1 * e^3 b_1$:
 $+|22aaaa00| - 2 \cdot |2aa2a00a| + 2 \cdot |2a2aa0a0| + 2 \cdot |2aa20aa0| - 2 \cdot |2a2a0a0a| - 2 \cdot |a2a2a0a0|$
 $+2 \cdot |a22aa00a| + 2 \cdot |a2a20a0a| - 2 \cdot |a22a0aa0| + |aa2200aa|$